

# Ethanol Blending in Gasoline - Guatemala

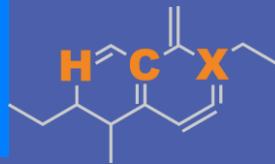
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# Ethanol Blending in Latin America

There are important fuel quality and environmental impact of vehicle emission challenges in the Region.

- The use of ethanol improves gasoline quality and creates flexibility in gasoline production.
- Ethanol use is a cost-effective way to increase gasoline octane and to replace more expensive gasoline components.
- Ethanol contributes to transport decarbonization and air quality improvement.
- There are opportunities across Latin America to increase the ethanol blend level and implement new policies on the use of gasoline-ethanol blends.

Sixteen countries with potential and additional use of ethanol were studied: 1) gasoline market profiles; 2) Optimization of gasoline blends with ethanol and 3) Environmental impact of gasolines blended with ethanol.



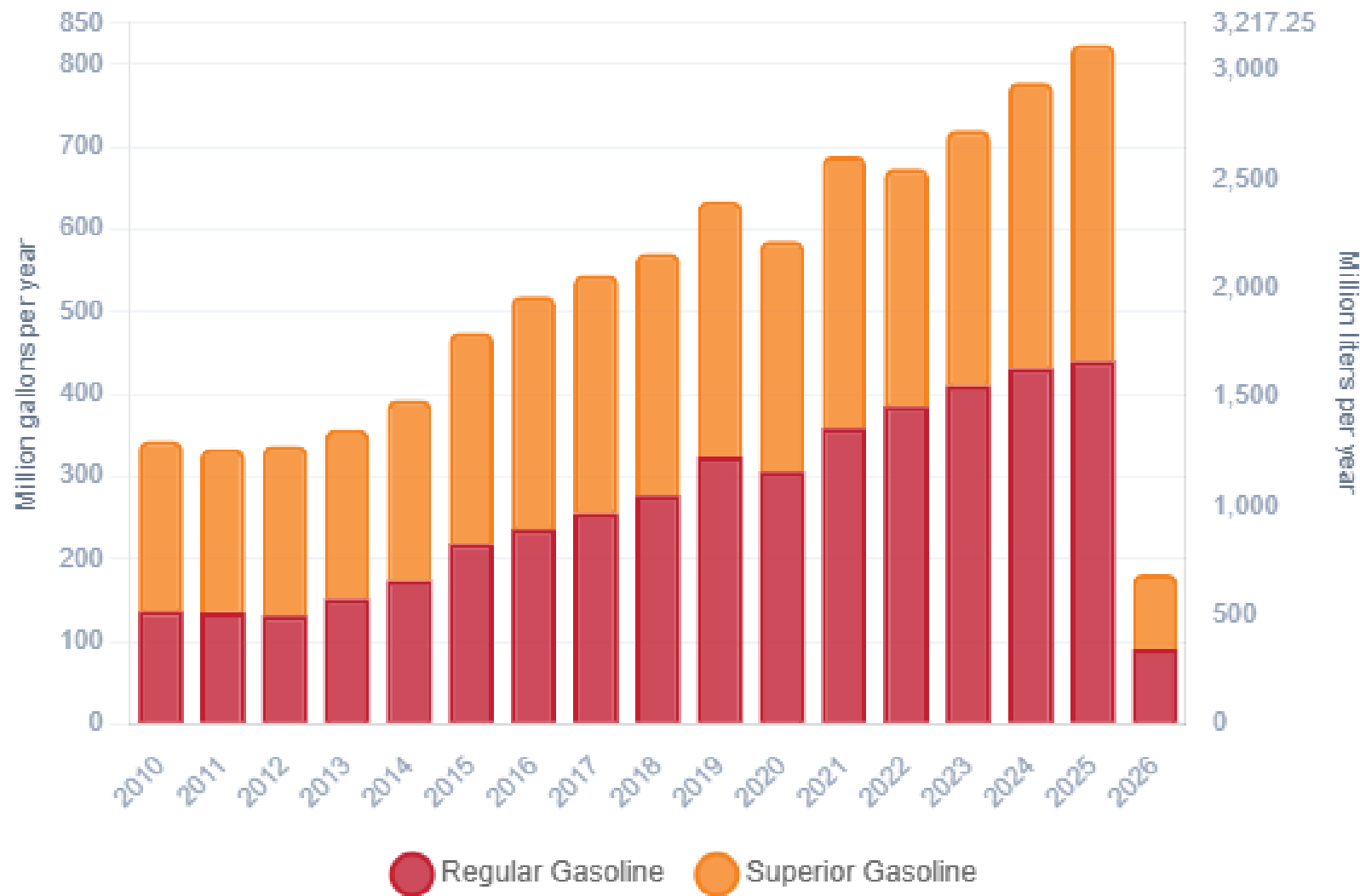
# Ethanol Blending in Gasoline - Guatemala



In 2025, gasoline consumption was 824 million gallons (3,119 million liters). Gasoline is supplied only by imports, mainly from United States. There are two grades of gasoline: (RON 91 - AKI 85) and Superior (RON 95 - AKI 89). 53% of gasoline is gasoline Regular.

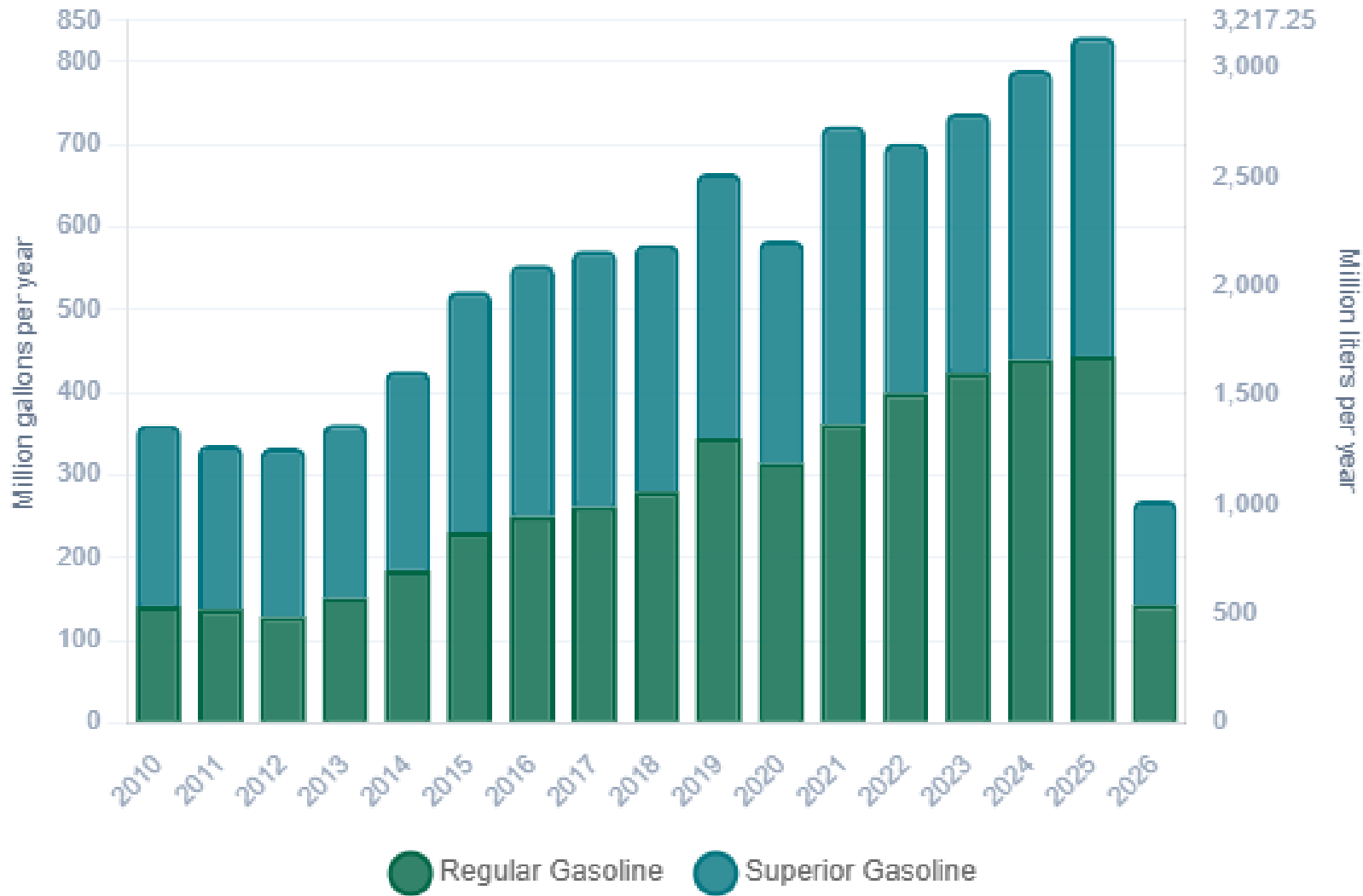
Guatemala approved the mandate for E10 blends throughout the territory in 2025. It has local ethanol advanced ethanol production that is exported to Europe. Guatemala is the main ethanol producer in Central America.

# Gasoline Demand in Guatemala



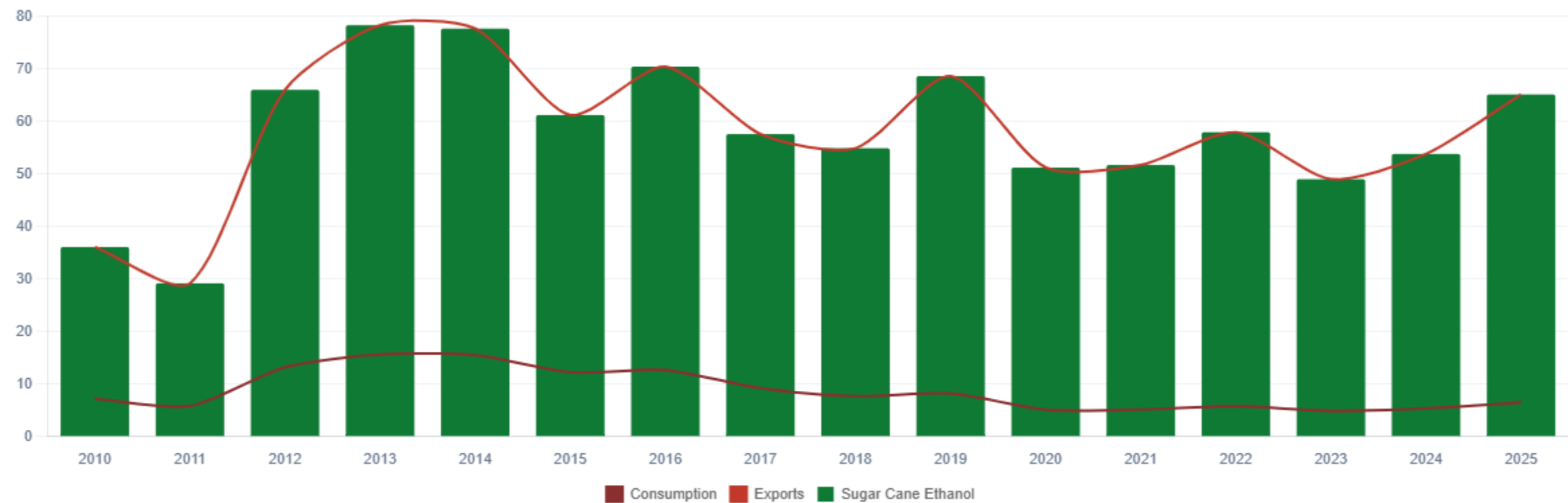
Source: MEM, 2025

# Gasoline Imports in Guatemala



Source: MEM, 2025

# Ethanol Balance in Guatemala



Source: Asociación de productores de etanol de Guatemala

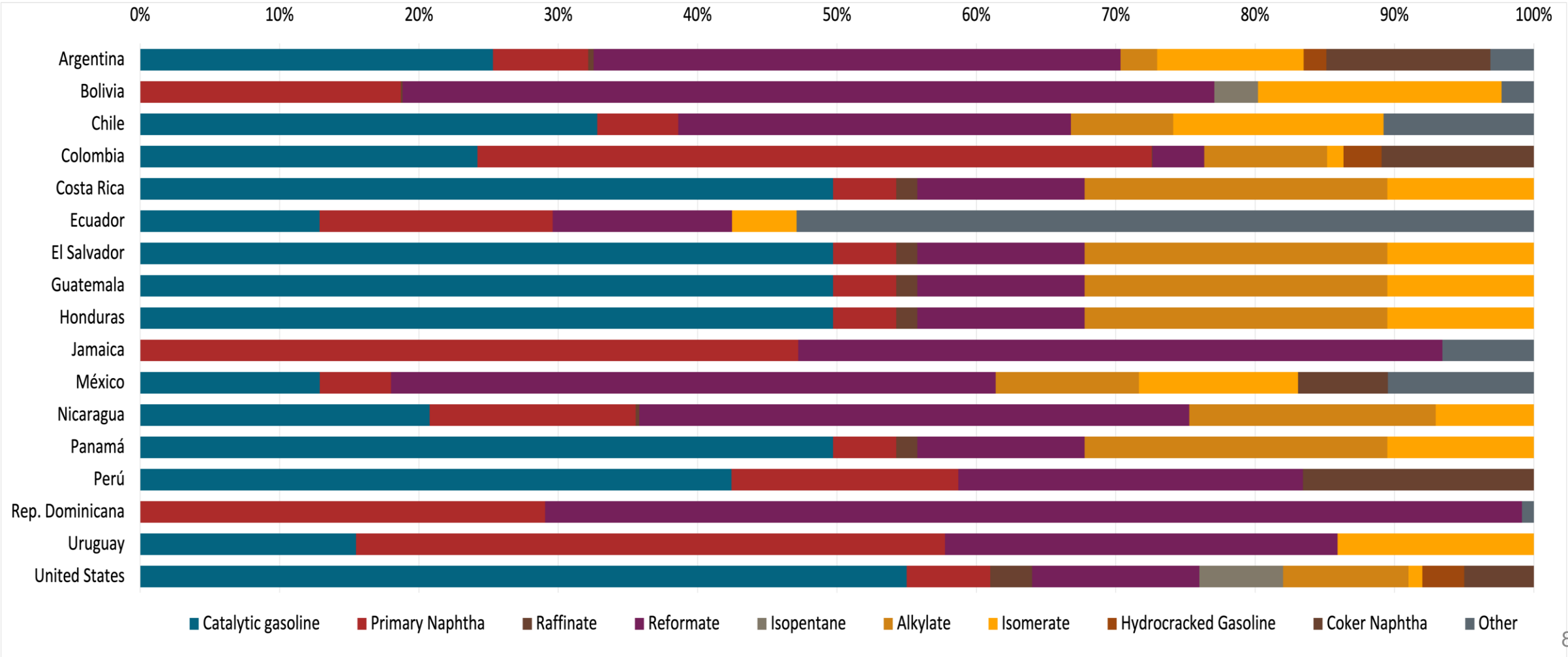
# Gasoline Quality in Guatemala

| Name                     | Ministerial Accord Number 320-2022 |                   | EN 228:2012 + A1:2017 (Euro 6 enabling)                                                   |            |                         |            |
|--------------------------|------------------------------------|-------------------|-------------------------------------------------------------------------------------------|------------|-------------------------|------------|
| Implementation Date      | 2022                               |                   | 2017                                                                                      |            |                         |            |
| Applicability            | Whole country                      | Whole country     | All countries                                                                             |            |                         |            |
| Selected Grade           | Gasoline Regular                   | Gasoline Superior | RON 95 E5                                                                                 | RON 95 E10 | RON 98 E5               | RON 98 E10 |
| Benzene Content          | < 2,5% v/v                         | < 2,5% v/v        | < 1 %v/v                                                                                  | < 1 %v/v   | < 1 %v/v                | < 1 %v/v   |
| Aromatics                | - / < 50% v/v                      | - / < 50% v/v     | < 35 %v/v                                                                                 | < 35 %v/v  | < 35 %v/v               | < 35 %v/v  |
| Olefins                  | - / < 30% v/v                      | - / < 30% v/v     | < 18 %v/v                                                                                 | < 18 %v/v  | < 18 %v/v               | < 18 %v/v  |
| Lead Content             | < 0,013 g/l                        | < 0,013 g/l       | < 5 mg/l                                                                                  | < 5 mg/l   | < 5 mg/l                | < 5 mg/l   |
| Manganese                | < 2,5 mg/l                         | < 2,5 mg/l        | < 2,0 mg/l                                                                                | < 2,0 mg/l | < 2,0 mg/l              | < 2,0 mg/l |
| RON                      | > 91                               | > 95              | > 95                                                                                      | > 95       | > 98                    | > 98       |
| MON                      | -                                  | -                 | > 85                                                                                      | > 88       | > 85                    | > 88       |
| AKI                      |                                    |                   |                                                                                           |            |                         |            |
| Sulfur Content           | < 500 mg/kg                        | < 500 mg/kg       | < 10 mg/kg                                                                                | < 10 mg/kg | < 10 mg/kg              | < 10 mg/kg |
| Oxygen Content           | -                                  | -                 | <2,7 % m/m                                                                                | <3,7 % m/m | <2,7 % m/m              | <3,7 % m/m |
| Ethanol (EtOH)           | < 10 %v/v                          | < 10 %v/v         | <5 %v/v                                                                                   | <10 %v/v   | <5 %v/v                 | <10 %v/v   |
| RVP 37.8°C (Summer)      | < 69 kPa                           | < 69 kPa          | <> 60 - 70 kPa *Depends on the country, RVP is regulated in the EU Fuel Quality Directive |            |                         |            |
| RVP 37.8 °C(Winter)      |                                    |                   |                                                                                           |            |                         |            |
| RVP 37.8°C (Transition)  |                                    |                   |                                                                                           |            |                         |            |
| MTBE                     | 10% v/v                            | 10% v/v           | -                                                                                         | -          | -                       | -          |
| Ethers 5 or more C Atoms | -                                  | -                 | Based on oxygen content                                                                   | <22 %v/v   | Based on oxygen content | <22 %v/v   |

Source: MEM, 2022

# Gasoline Component Blending in Latin America

Gasoline is a blend of a base gasoline and other components. This blending is usually done at blending terminals as only 30% of the world's finished gasoline is distributed directly from refineries. Each component provides different properties to the final blend, for example, isomerates, alkylates and butanes increase the octane. The components commonly used in Latin America are:





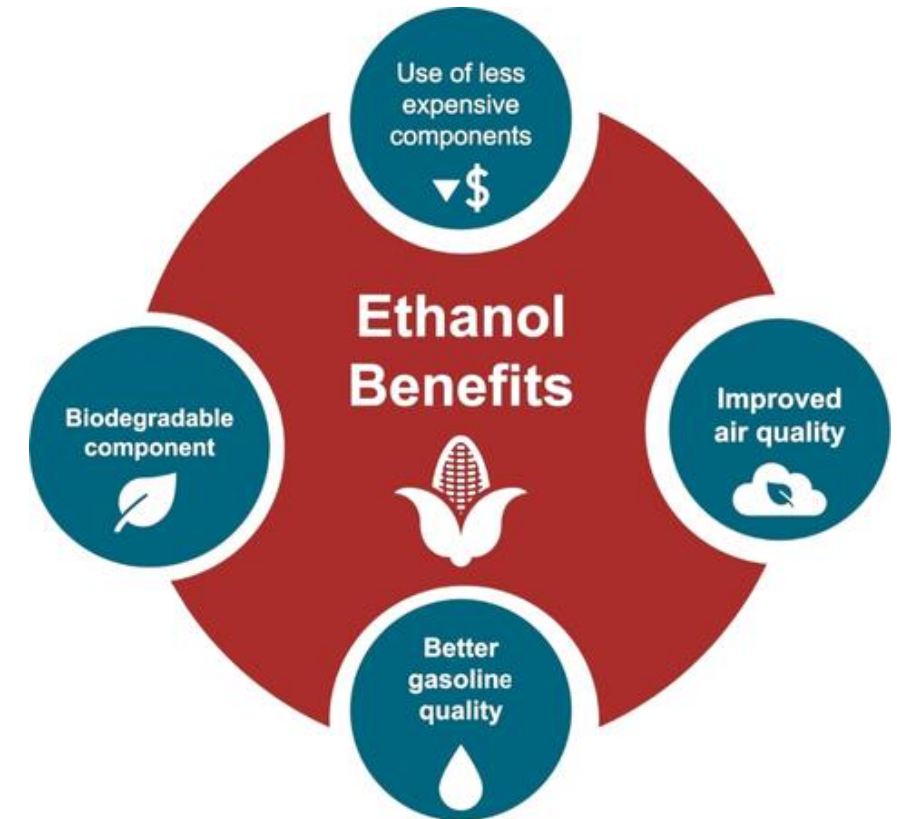
# Gasoline Blending Optimization

In some parts of the world, ethanol is added to gasoline as a blending component. The advantages of ethanol include that it is a renewable fuel made of biomass; that it is an octane booster that helps to dilute sulfur; and that it allows the fulfillment of environmental objectives. To determine the optimal components to be blended with ethanol, a **blending model** was used. This model selects the components to add in the gasoline/ethanol blend based on:

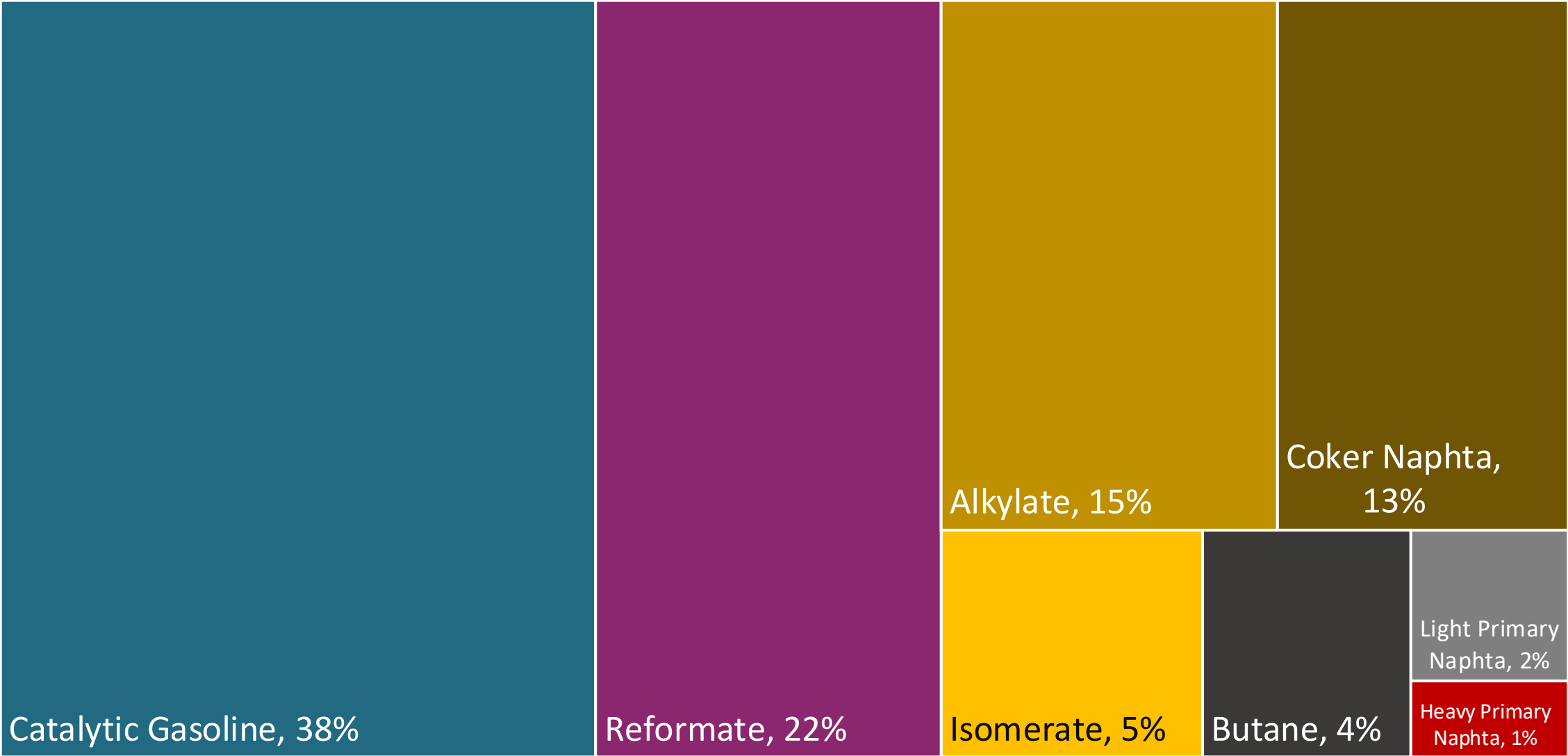
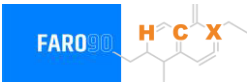
- Components prices,
- Properties each component affects,
- Quality parameters by country, and
- Component availability by country.

Through iterations, the model obtains the %v/v of the components to be blended with 10%, 15%, 20%, 25% and 30% of ethanol, in such a way that the final blend complies with the required properties of a finished gasoline by country.

The blending model uses gasoline component spot average prices 2025 and provides fuel prices that do not include country distribution costs, local taxes and subsidies and import or gas station margins.

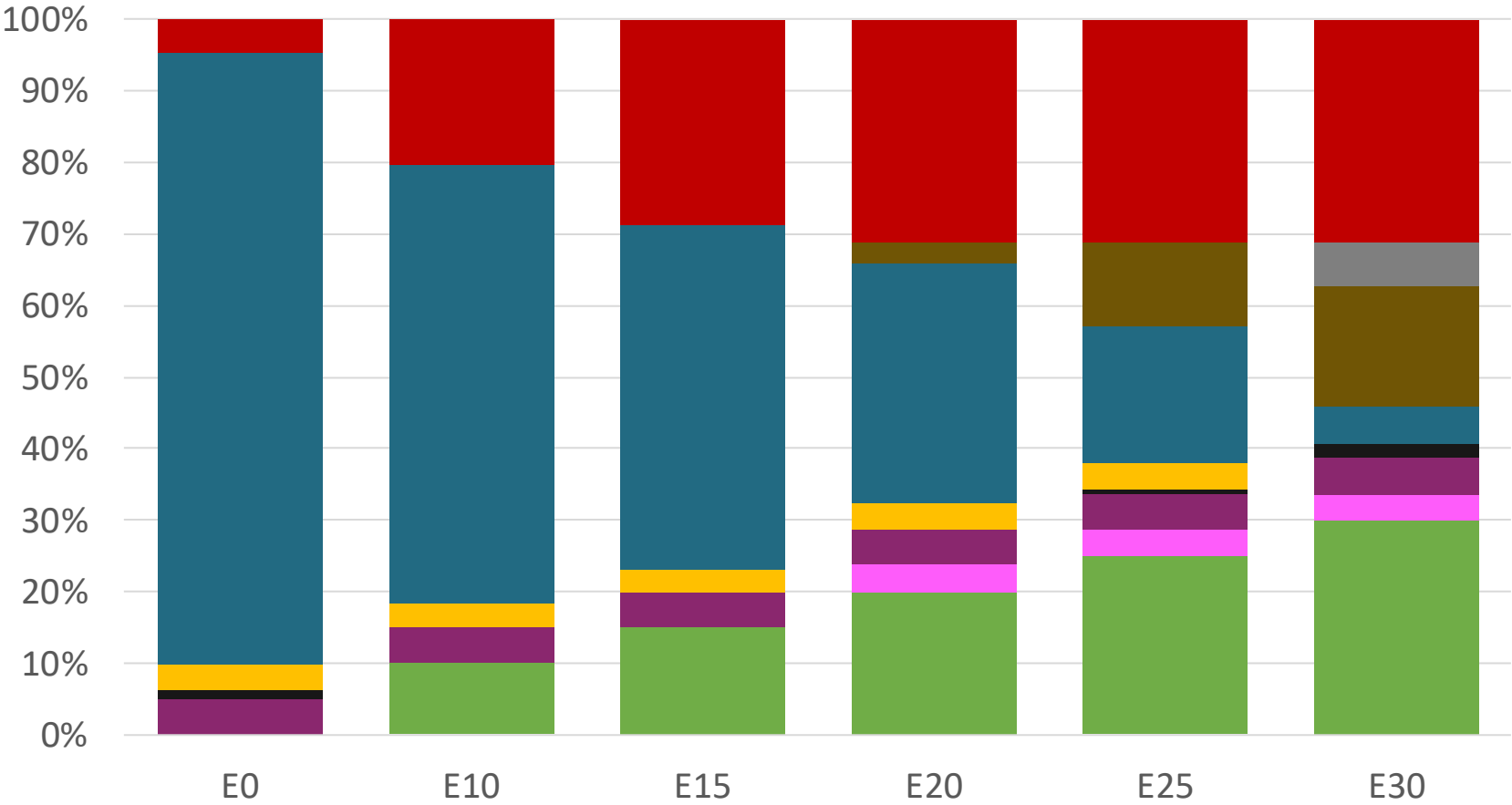
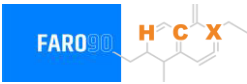


# Blending Components - Guatemala



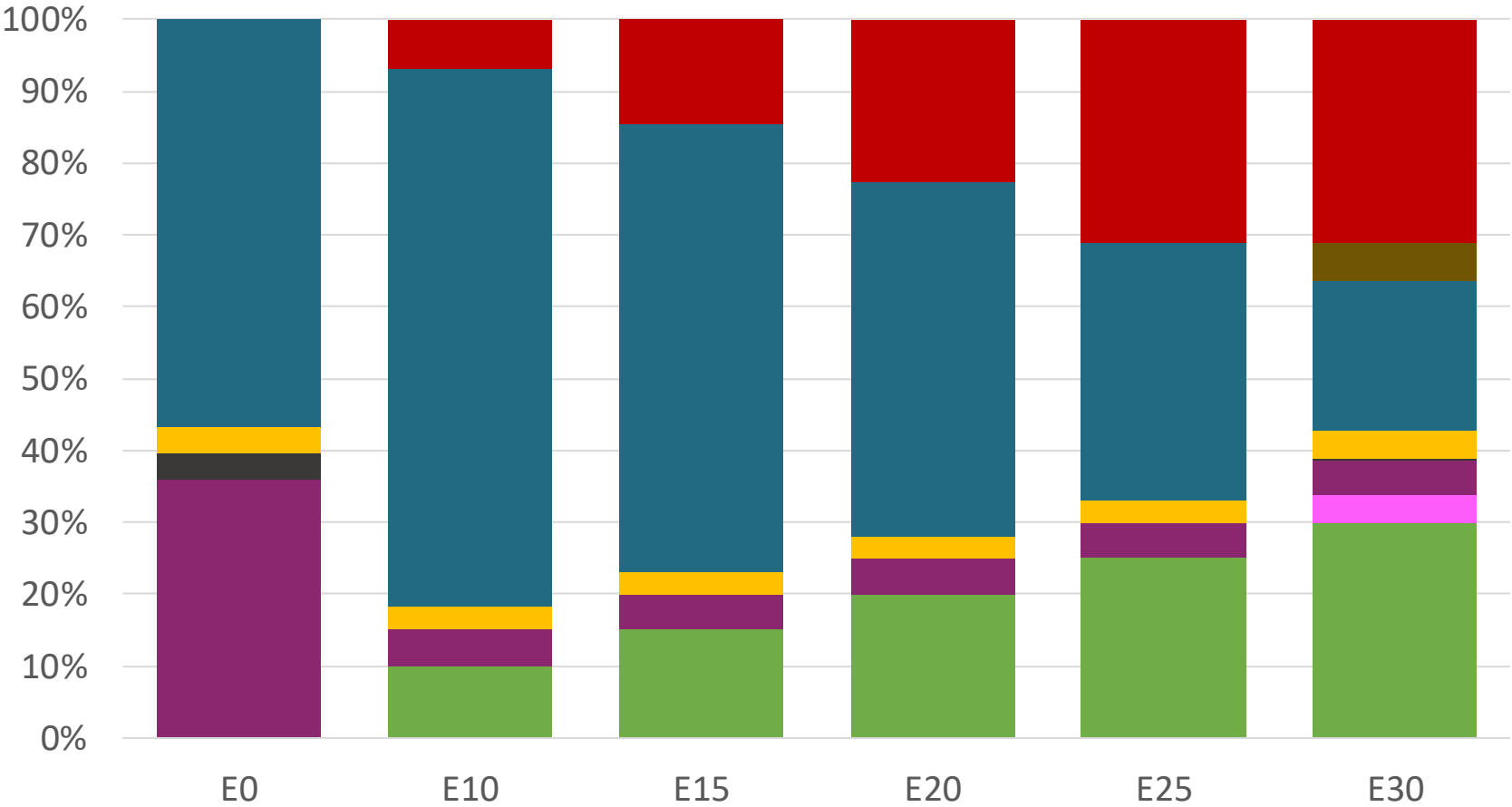
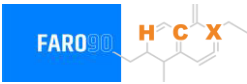
Source: Faro90

# Guatemala – Regular – Constant Octane



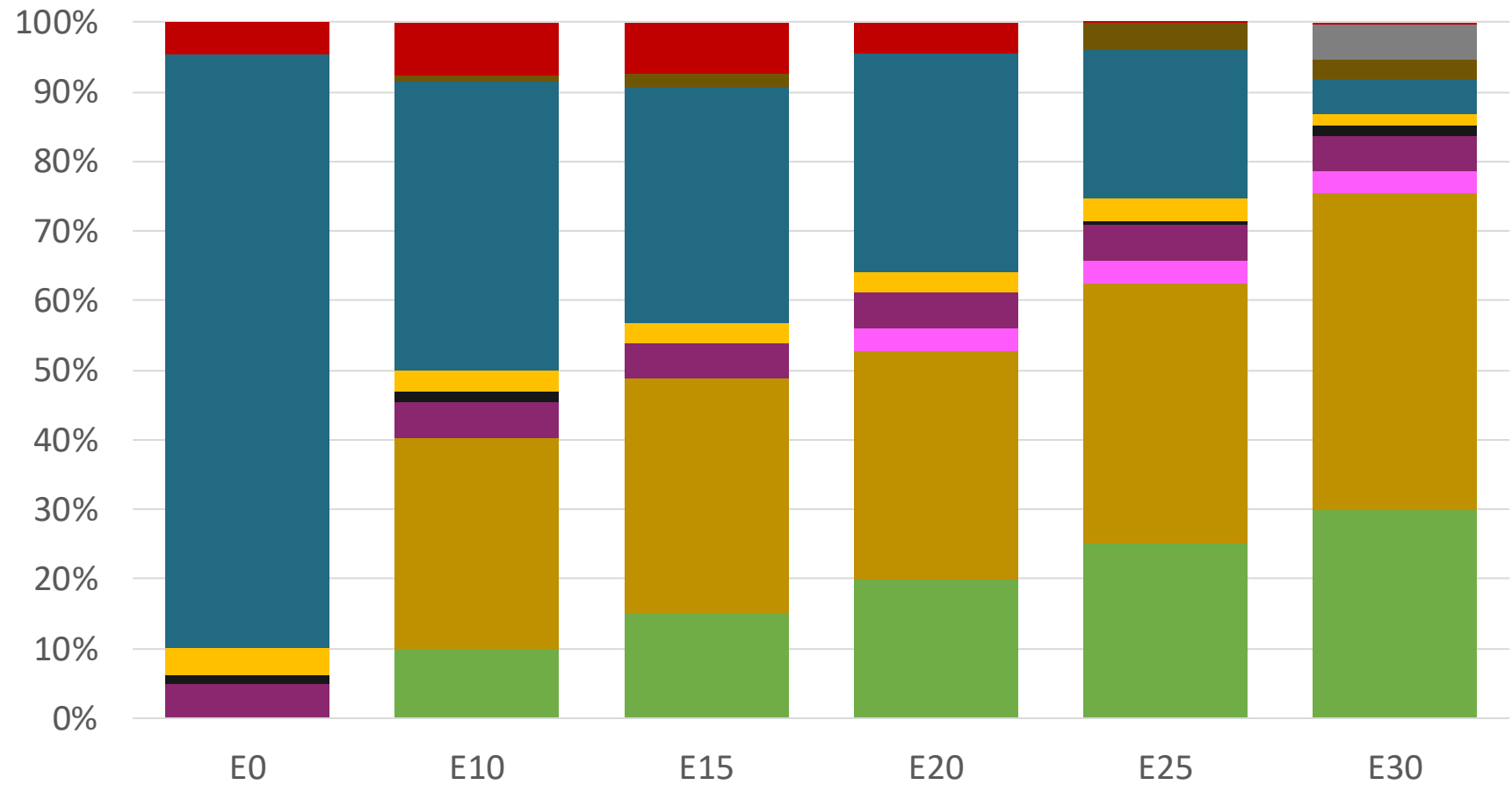
|                 |          |           |           |           |           |           |
|-----------------|----------|-----------|-----------|-----------|-----------|-----------|
| Octane (RON)    | 91.0     | 91.0      | 91.0      | 91.0      | 91.0      | 91.0      |
| Price (USD/gal) | \$ 2.126 | \$. 1.867 | \$. 1.723 | \$. 1.576 | \$. 1.425 | \$. 1.307 |

# Guatemala – Premium – Constant Octane



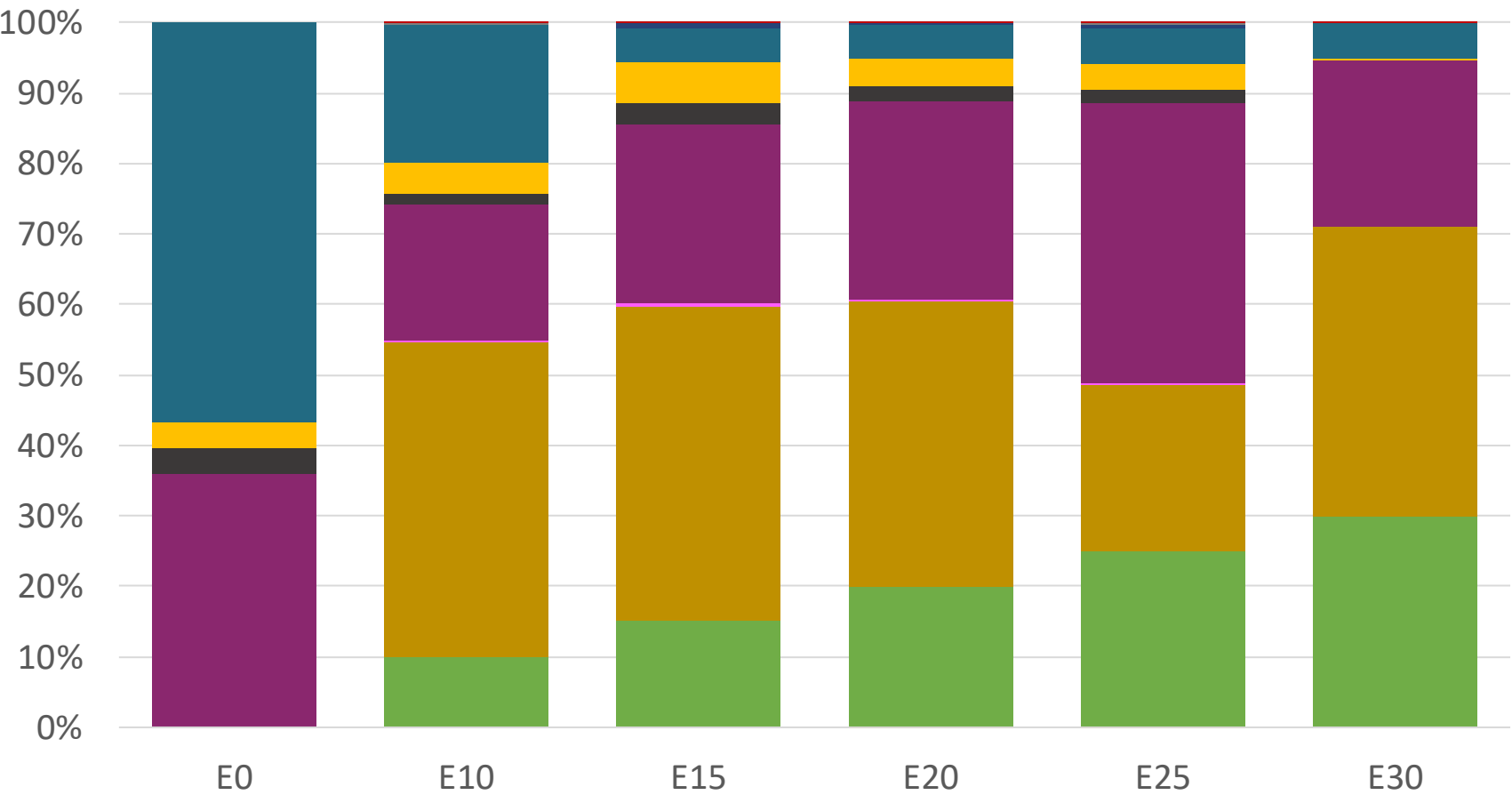
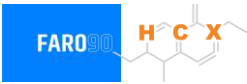
|                 |          |          |          |          |          |          |
|-----------------|----------|----------|----------|----------|----------|----------|
| Octane (RON)    | 95.0     | 95.0     | 95.0     | 95.0     | 95.0     | 95.0     |
| Price (USD/gal) | \$ 2.376 | \$ 2.073 | \$ 1.938 | \$ 1.796 | \$ 1.647 | \$ 1.498 |

# Guatemala – Regular – Increased Octane



|                 |          |          |          |          |          |          |
|-----------------|----------|----------|----------|----------|----------|----------|
| Octane (RON)    | 91.0     | 94.6     | 96.6     | 99.1     | 101.2    | 102.4    |
| Price (USD/gal) | \$ 2.126 | \$ 2.126 | \$ 2.126 | \$ 2.126 | \$ 2.126 | \$ 2.126 |

# Guatemala – Premium – Increased Octane



|                 |          |          |          |          |          |          |
|-----------------|----------|----------|----------|----------|----------|----------|
| Octane (RON)    | 95.0     | 98.0     | 100.3    | 102.6    | 104.9    | 105.8    |
| Price (USD/gal) | \$ 2.376 | \$ 2.376 | \$ 2.376 | \$ 2.376 | \$ 2.376 | \$ 2.376 |

# Vehicle Emission Impact for Ethanol Gasoline Blending

The model used in this analysis takes as a reference the [International Vehicle Emissions Model \(IVE\)](#).

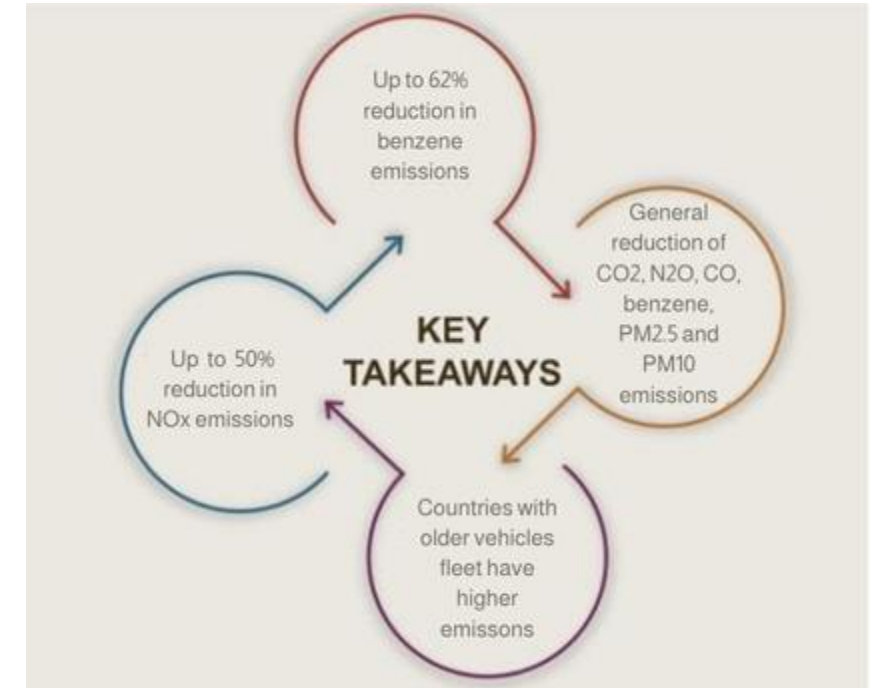
The model uses the Base Emission Rates from IVE model, as well as its Adjustment Factors based on:

- Vehicle technology (cars, trucks, buses, motorcycles),
- Vehicle fleet average age,
- Average traveled distance per vehicle by country, as well as
- Geographical and climatic conditions (altitude, humidity, temperature).

Emissions of criteria pollutants, toxic pollutants, and greenhouse gases (GHG) were calculated and calibrated with emission inventories, using real gasoline quality data. The reduction rates for gasoline/ethanol blends were obtained from various sources (IPCC, US Grains, among others).

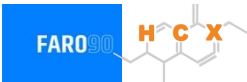
Emission estimations for different pollutants for gasoline and gasoline/ethanol blends (10%, 15%, 20%, 25% and 30% ethanol) were determined using the IVE Model. A comparison between the results and the European (Euro 6) requirements is made. Results are also compared with real emissions of the United States vehicle fleet\*.

## Main Results



\*Source: Bureau of transportation statistics.

# Gasoline Vehicular Fleet - Guatemala



| Vehicle type       | Engine       | Exhaust             | Recovery                     | Age                           | Number of vehicles              |                              |  |  |
|--------------------|--------------|---------------------|------------------------------|-------------------------------|---------------------------------|------------------------------|--|--|
| Light automobiles  | Carburator   | 3-Ways              | PCV Tank                     | > 8 years, > 160 tkm          | <div><div>14,668</div></div>    |                              |  |  |
|                    | Single FI-Pt | 3-Ways              |                              | > 8 years, > 160 tkm          | <div><div>51,436</div></div>    |                              |  |  |
|                    | Multi FI-Pt  | 3-Ways              |                              | > 8 years, > 160 tkm          | <div><div>362,121</div></div>   |                              |  |  |
|                    |              | Euro III            |                              | > 8 years, > 160 tkm          | <div><div>239,410</div></div>   |                              |  |  |
|                    |              | Euro IV             |                              | > 8 years, > 160 tkm          | <div><div>160,262</div></div>   |                              |  |  |
|                    |              | Euro V / Euro 6     |                              | > 8 years, > 160 tkm          | <div><div>97,733</div></div>    |                              |  |  |
|                    |              |                     |                              | 4-8 years, 80-160 tkm         | <div><div>52,603</div></div>    |                              |  |  |
|                    |              | < 4 years, < 80 tkm |                              | <div><div>27,177</div></div>  |                                 |                              |  |  |
|                    |              | Hybrid              |                              |                               | Any                             | <div><div>6,278</div></div>  |  |  |
|                    | Electric     |                     |                              | Any                           | <div><div>785</div></div>       |                              |  |  |
| Medium automobiles | Carburator   | 3-Ways              | PCV Tank                     | > 8 years, > 160 tkm          | <div><div>33,594</div></div>    |                              |  |  |
|                    | Single FI-Pt | 3-Ways              |                              | > 8 years, > 160 tkm          | <div><div>294,887</div></div>   |                              |  |  |
|                    | Multi FI-Pt  | 3-Ways              |                              | > 8 years, > 160 tkm          | <div><div>650,178</div></div>   |                              |  |  |
|                    |              | Euro III            |                              | > 8 years, > 160 tkm          | <div><div>277,862</div></div>   |                              |  |  |
|                    |              | Euro IV             |                              | > 8 years, > 160 tkm          | <div><div>160,947</div></div>   |                              |  |  |
|                    |              | Euro V / Euro 6     |                              | > 8 years, > 160 tkm          | <div><div>120,025</div></div>   |                              |  |  |
|                    |              |                     |                              | 4-8 years, 80-160 tkm         | <div><div>116,456</div></div>   |                              |  |  |
|                    |              | < 4 years, < 80 tkm |                              | <div><div>165,761</div></div> |                                 |                              |  |  |
|                    |              | Hybrid              |                              |                               | Any                             | <div><div>11,362</div></div> |  |  |
|                    |              | Electric            |                              |                               | Any                             | <div><div>1,420</div></div>  |  |  |
| Small Engines      | FI 4-cycles  | None                | PCV                          | > 8 years, > 160 tkm          | <div><div>49,306</div></div>    |                              |  |  |
|                    |              | Euro II             |                              | > 8 years, > 160 tkm          | <div><div>59,976</div></div>    |                              |  |  |
|                    |              | Euro III            |                              | > 8 years, > 160 tkm          | <div><div>645,317</div></div>   |                              |  |  |
|                    |              | Euro IV             |                              | > 8 years, > 160 tkm          | <div><div>141,330</div></div>   |                              |  |  |
|                    |              | Euro V              |                              | 4-8 years, 80-160 tkm         | <div><div>483,674</div></div>   |                              |  |  |
|                    |              |                     |                              | 4-8 years, 80-160 tkm         | <div><div>200,576</div></div>   |                              |  |  |
|                    |              |                     |                              | < 4 years, < 80 tkm           | <div><div>1,442,641</div></div> |                              |  |  |
|                    |              | Electric            |                              |                               | Any                             | <div><div>974</div></div>    |  |  |
|                    | Heavy        | Carburator          | 3-Ways                       | PCV                           | > 8 years, > 160 tkm            | <div><div>16,289</div></div> |  |  |
| FI                 |              | 3-Ways              | > 8 years, > 160 tkm         |                               | <div><div>232,495</div></div>   |                              |  |  |
|                    |              | Euro III            | > 8 years, > 160 tkm         |                               | <div><div>74,845</div></div>    |                              |  |  |
|                    |              | Euro IV             | > 8 years, > 160 tkm         |                               | <div><div>36,066</div></div>    |                              |  |  |
|                    |              | Euro V / Euro 6     | > 8 years, > 160 tkm         |                               | <div><div>20,853</div></div>    |                              |  |  |
|                    |              |                     | 4-8 years, 80-160 tkm        |                               | <div><div>27,307</div></div>    |                              |  |  |
|                    |              | < 4 years, < 80 tkm | <div><div>52,267</div></div> |                               |                                 |                              |  |  |
|                    |              | Hybrid              |                              |                               | Any                             | <div><div>9</div></div>      |  |  |
|                    |              | Electric            |                              |                               | Any                             | <div><div>215</div></div>    |  |  |

Gasoline Vehicular Fleet: 5,915,613

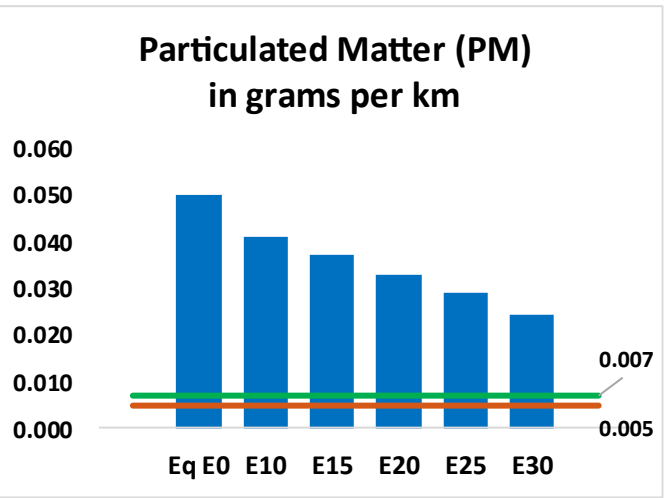
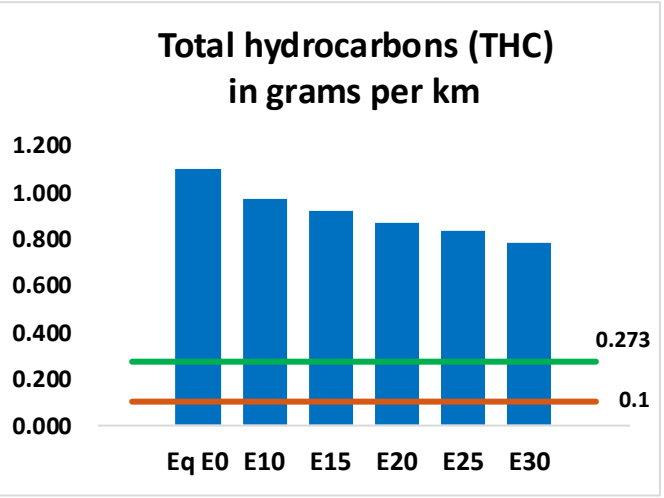
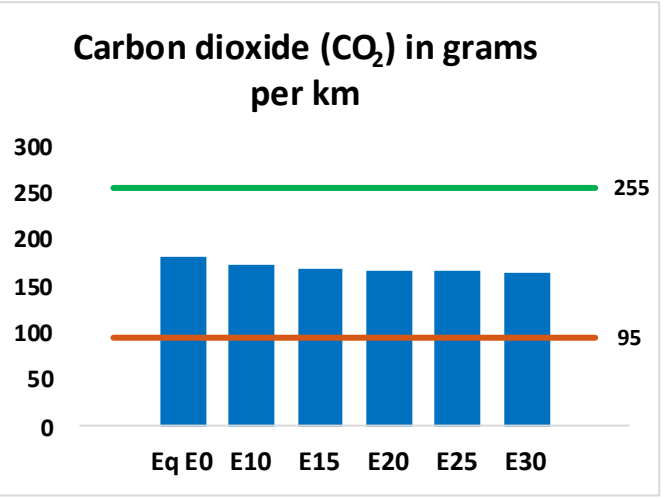
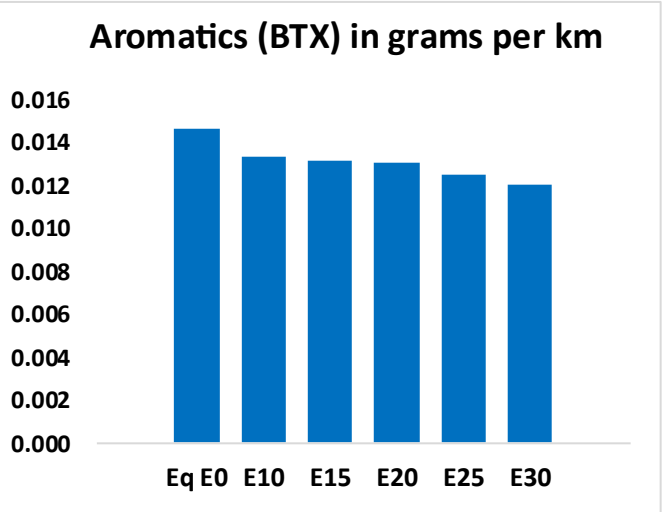
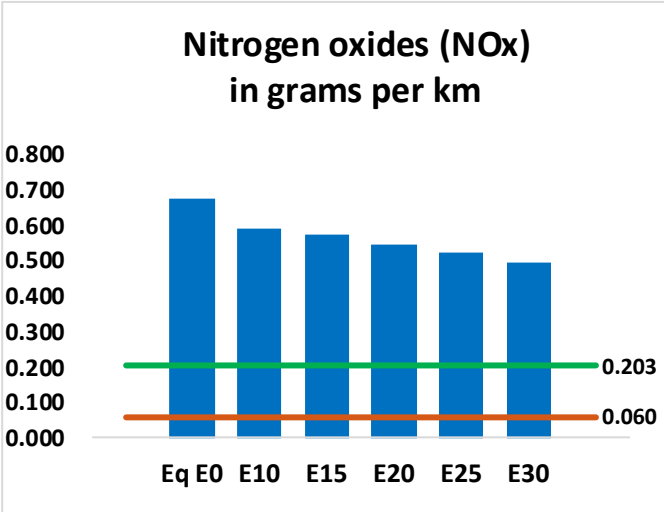
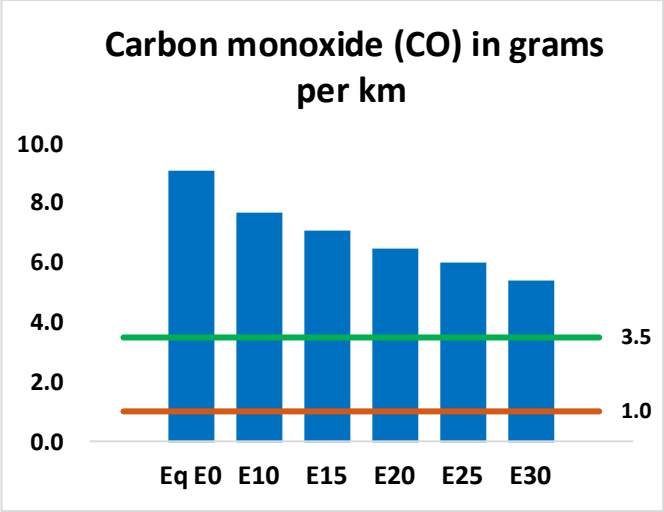
Average age: 13 years

Motorcycles: 48%

Source: INE, analysis Faro 90



# Guatemala – Vehicular Emissions



# Guatemala – Vehicular Emissions

| Vehicular Fleet Emissions (kg/day) | Eq E0      | E5         | E10        | E15        | E20        | E25        | E30        | Reduction (%) |      |
|------------------------------------|------------|------------|------------|------------|------------|------------|------------|---------------|------|
|                                    |            |            |            |            |            |            |            | E10           | E20  |
| CO                                 | 964,717    | 887,827    | 810,937    | 747,397    | 683,210    | 633,780    | 572,436    | -16%          | -29% |
| CO2                                | 19,217,409 | 18,723,247 | 18,256,539 | 17,889,486 | 17,707,854 | 17,630,789 | 17,305,803 | -5%           | -8%  |
| VOC                                | 115,959    | 109,110    | 102,260    | 97,050     | 91,993     | 88,160     | 82,847     | -12%          | -21% |
| NOx                                | 71,550     | 66,064     | 62,410     | 60,606     | 57,636     | 55,337     | 52,612     | -13%          | -19% |
| SOx                                | 220        | 203        | 186        | 172        | 159        | 144        | 129        | -15%          | -28% |
| NH3                                | 7,058      | 6,988      | 6,918      | 6,951      | 7,029      | 7,084      | 7,130      | -2%           | 0%   |
| N2O                                | 524        | 522        | 520        | 522        | 533        | 539        | 545        | -1%           | 2%   |
| CH4                                | 26,581     | 26,581     | 26,581     | 26,980     | 27,565     | 28,070     | 28,548     | 0%            | 4%   |
| Lead                               | -          | -          | -          | -          | -          | -          | -          | 0%            | 0%   |
| Butadiene                          | 951        | 905        | 859        | 827        | 797        | 774        | 742        | -10%          | -16% |
| Acetaldehyde                       | 2,182      | 2,504      | 3,675      | 5,596      | 7,624      | 8,712      | 10,294     | 68%           | 249% |
| Formaldehyde                       | 7,888      | 7,669      | 8,940      | 10,497     | 10,998     | 12,166     | 13,244     | 13%           | 39%  |
| Benzene                            | 1,554      | 1,493      | 1,415      | 1,392      | 1,381      | 1,320      | 1,278      | -9%           | -11% |
| PM2.5                              | 1,102      | 1,002      | 902        | 813        | 727        | 635        | 538        | -18%          | -34% |
| PM10                               | 4,202      | 3,820      | 3,438      | 3,102      | 2,773      | 2,420      | 2,051      | -18%          | -34% |

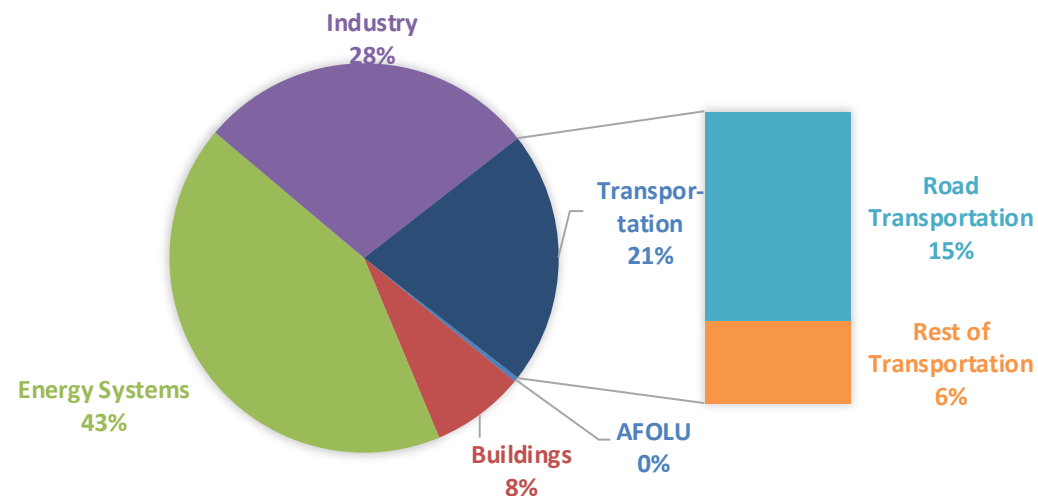
# Life Cycle GHG Emissions

The Global Agenda for Climate Action calls for countries, cities and regions, businesses, and members of civil society around the world to achieve a low-carbon and climate-resilient economies in support of the Paris Agreement. Road Transportation accounts for 15% of total GHG emissions (IPCC,2023).

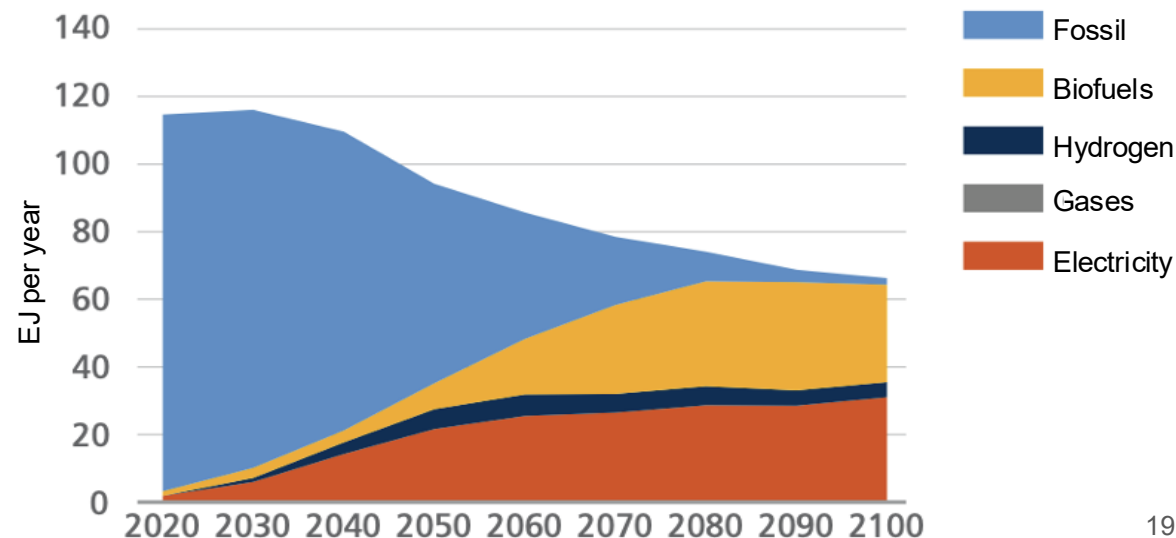
Ethanol gasoline blends are an internationally adopted practice that offer significant short-term benefits. Biofuels are an important solution to achieve reduction targets in the transportation sector.

Country GHG emission reductions are calculated using **RED II/III** and **GREET** life cycle emission methodologies and the **International Vehicular Emissions Model (IVE)**. Results are compared to the current country targets for 2035. Current 2025 country emission are those published by **IPCC**.

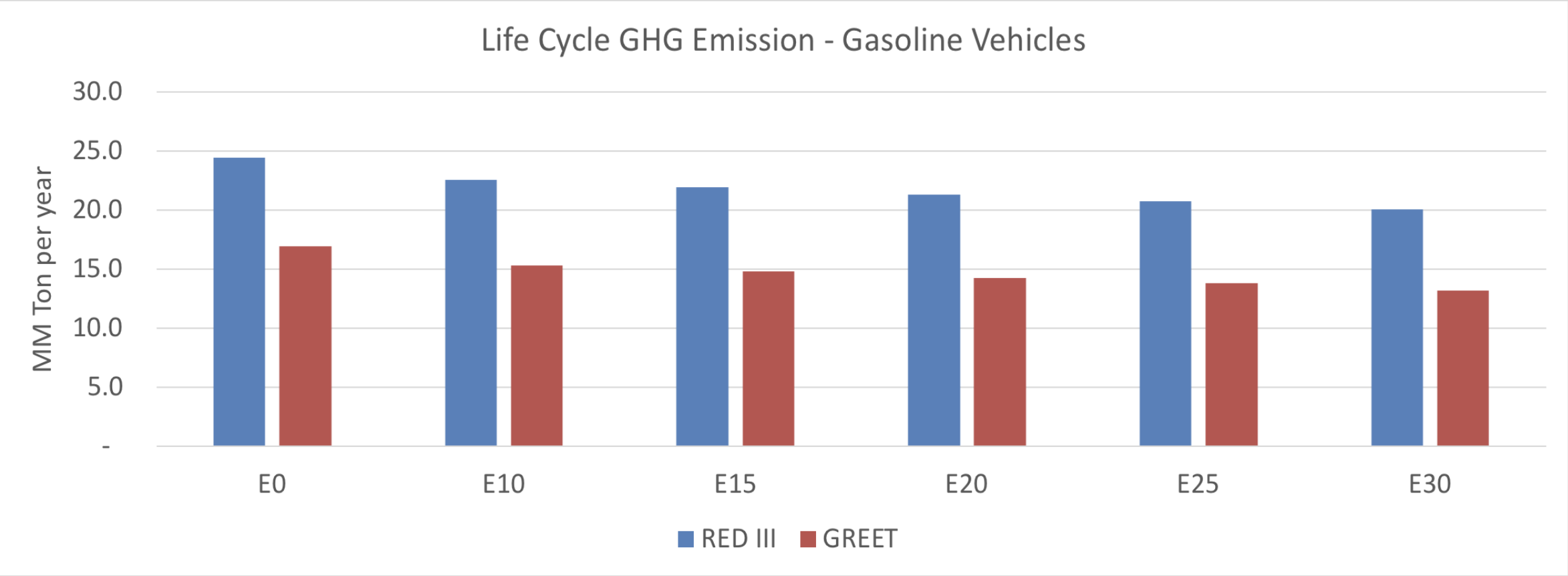
Current GHG Global Emissions Distribution



IPCC Transport Sector Sustainable Development Goals and climate policies Scenario, 2023



# Guatemala – Gasoline Vehicle Fleet GEI Emissions



|                                   |      |
|-----------------------------------|------|
| Country Emissions 2025 (MMT/year) | 43.5 |
| 2035 Target (MMT/year)            | 21.7 |
| Delta                             | 21.7 |

|                  |      |      |       |       |       |       |
|------------------|------|------|-------|-------|-------|-------|
| %Reduction vs E0 | E0   | E10  | E15   | E20   | E25   | E30   |
| GREET 2025       | 0.0% | 9.8% | 12.7% | 15.9% | 18.6% | 22.1% |
| RED II/III       | 0.0% | 7.7% | 10.1% | 12.8% | 15.0% | 17.8% |

|               |      |      |       |       |       |       |
|---------------|------|------|-------|-------|-------|-------|
| % 2035 Target | E0   | E10  | E15   | E20   | E25   | E30   |
| GREET 2025    | 0.0% | 7.6% | 9.9%  | 12.4% | 14.5% | 17.2% |
| RED II/III    | 0.0% | 8.6% | 11.4% | 14.4% | 16.9% | 20.0% |

Source: Greet, RED II/III, climateactiontracker.org, F90